

WASP-36b

R-1718

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_210404 · created 2026-08-02T16:14:17+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459308.7042 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.101 +/- 0.035
Transit depth (Rp/Rs)^2	1.02 +/- 0.71 [%]
Orbital Inclination (inc)	86.85 +/- 2.62
Airmass coefficient 1 (a1)	1.033 +/- 0.02
Airmass coefficient 2 (a2)	-0.0193 +/- 0.0091
Scatter in the residuals of the lightcurve fit is	1.92 %
Best Comparison Star	#2 - [437, 42]
Optimal Method	PSF photometry
Transit Duration (day)	0.0858 +/- 0.0092

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.101 ± 0.035 vs 0.1368 (deviation 0.69σ)
Duration fit vs published	0.0858 d vs 0.0758 d (deviation 0.73σ)
Mid-time O-C	-9.8 min
Depth SNR	2.0
Residual scatter	1.92 %

Light curve

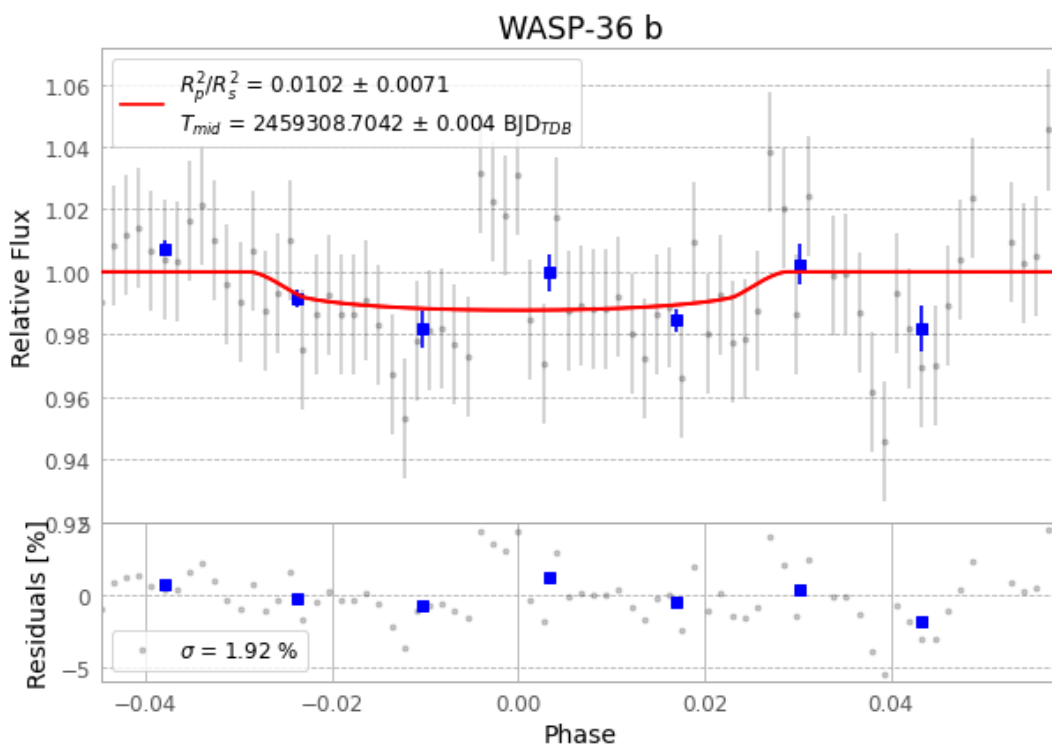


Figure 1 — FinalLightCurve_WASP-36 b_2021-04-03.png (SHA-256 18cdc398f393612e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 73], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6a9ac0597b2e60a5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-36210404030912.FITS → WASP-36210404065712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-210404.log (0008e58173e1...), FinalLightCurve_WASP-36 b_2021-04-03.png (18cdc398f393...), FinalLightCurve_WASP-36 b_2021-04-03.csv (608fb5ce469a...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 16:14Z.